for a network timeout. Applications will of course need to be updated to be Vista-aware, but commands such as “New view” already allow for cancellation of I/O requests (you can **[Ctrl] + [C]** a **net view** command if it seems unresponsive).

3.5 I/O Priority

While Vista makes allocations to throttle the CPU utilisation of a thread, it would be of little use if the I/O requests were not similarly prioritised and referenced. Say for example, Vista running several background applications such as a disk defragmenter, an anti-virus scanner that scans individual files in the background, an indexing service that indexes disk content for faster service—in this scenario, running a foreground task might bring the system to its knees if the background tasks weren’t properly prioritised. A foreground task should not have to wait for a background task to finish in order to access a disk. Vista thus introduces I/O prioritisation in order to ensure that foreground applications get preference.

Each class of service and application under Vista gets a priority assigned to it. Thus for example, the Memory Manager of the kernel uses the Critical priority when it wishes to flush data from RAM to the hard drive, in order to make space in the memory of the system. I/O in general has a default priority of Medium; priority for applications such as search indexing and Windows Defender scanning use Very Low I/O priority, while most storage devices are given Medium I/O priority.

Apart from I/O priorities, Vista also offers a bandwidth reservation system which ensures a quality of service (QoS) to applications, especially multimedia applications. Under this system, an application such as Windows Media Player, along with MMCSS priority boosts, can deliver nearly stutter-free playback of local content. Such an application would typically ask the I/O system to